INTRODUCTORY LECTURE

Cloud Computing

History, current market structure, and why cloud became the default environment for modern data science.

I. Origins and historical inflection points
II. Vendors, technologies, and real-world use
III. The cloud-native data science workflow

What do we mean by “cloud computing”?

A precise definition matters, because the cloud is more than “someone else’s server.”

“A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.”

— adapted from NIST SP 800-145

Five essential characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

I **History**
time-sharing to hyperscale

II **Current state**
vendors, platforms, technologies

III **Data science**
why the workflow moved to cloud

PART I

The History of the Cloud

From shared mainframes to the modern idea of computing as a programmable utility.

The cloud did not appear suddenly; it emerged from several older computing traditions.

Cloud computing is a convergence of older ideas

The key story is the gradual transformation of computing from owned hardware to remotely provisioned service.

The breakthrough was not simply "renting servers." It was combining shared computing, networks, virtualization, software-defined control, and metered economics into one operational model.

1950s–70s

Time-sharing + networked access

Shared mainframes and early networking establish the idea that computing can be a shared service.

1970s

Virtual machines

One physical system can host many isolated environments, increasing utilization.

2002–06

Commercial public cloud

AWS turns storage and compute into rentable, on-demand infrastructure.

2008–14

Cloud-native operations

Managed platforms, containers, Kubernetes, and Lambda reduce deployment friction.

2022–26

AI-era hyperscale

GenAI workloads push cloud growth, accelerators, and custom silicon.

Why the cloud finally worked

Earlier visions of utility computing became practical only when four conditions matured together.

Abstraction

Virtualization and containers turned hardware into software-defined pools of resources.

Programmability

APIs, infrastructure-as-code, and automation made remote infrastructure operable at scale.

Economics

Hyperscale data centers and pay-as-you-go pricing changed the cost model for capacity.

Operations

Managed services lowered the burden of patching, scaling, backup, and reliability engineering.

Elasticity alone was not enough. Cloud became compelling when elasticity, software control, global networks, and managed operations arrived together.

PART II

The Current State of Cloud

Today's cloud market is shaped by hyperscaler scale, cloud-native software, AI infrastructure, and real-world dependence across industries.

The cloud is now ordinary infrastructure for firms that need extraordinary scale.

The modern cloud spans several service and deployment models

Most organizations consume a mix of infrastructure, managed platforms, and software services at the same time.

Service layers

SaaS — Finished applications delivered over the web

Managed platform / PaaS — Databases, analytics, ML, serverless, Kubernetes

IaaS — Compute, storage, networking, accelerators

Deployment choices

Public cloud — Fastest path to elasticity, broad service catalogs, and global reach.

Hybrid cloud — Connect on-premises resources to cloud control planes for policy, identity, and data locality.

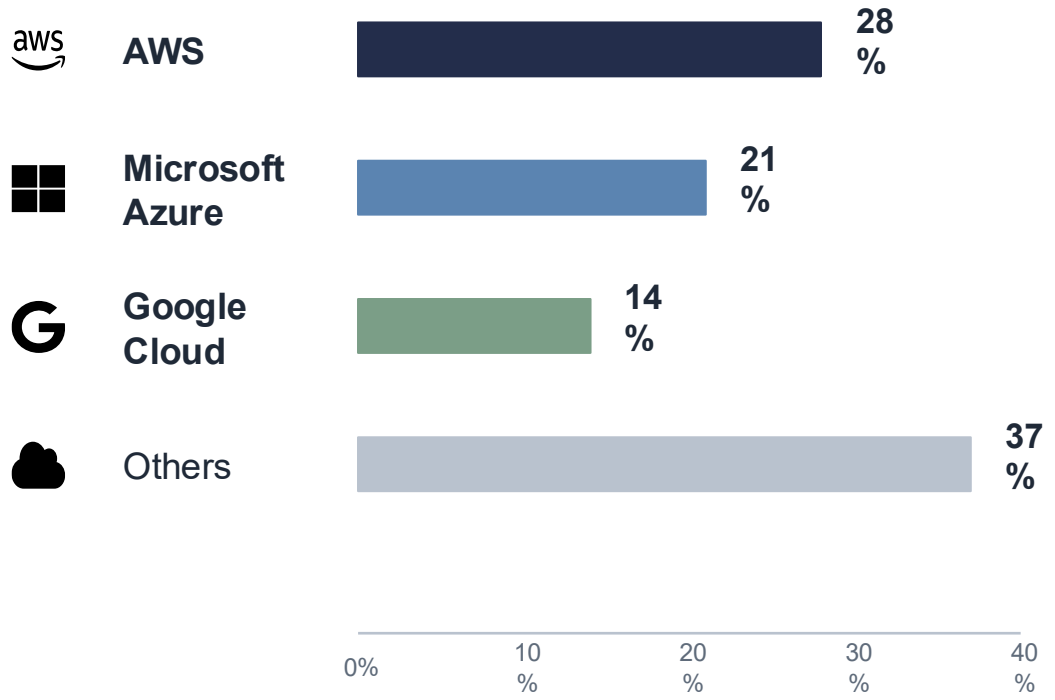
Multicloud — Use more than one provider selectively for resilience, regulation, bargaining power, or special capabilities.

Key point: enterprises rarely choose just one box on this slide; they combine layers and deployment models according to workload, regulation, and legacy constraints.

A few hyperscalers dominate infrastructure services

But competition is increasingly shaped by AI infrastructure, data platforms, and hybrid control rather than raw compute alone .

Global cloud infrastructure services revenue reached roughly $419B in 2025, with the top three providers still controlling most of the market.



$419B
estimated 2025 cloud infrastructure revenue

$119B
Q4 2025 revenue in one quarter

How providers differentiate now

- AWS: depth of services and ecosystem breadth
- Azure: enterprise identity, Microsoft integration, hybrid reach
- Google Cloud: analytics, data platforms, and AI/TPU strengths
- Specialists: edge, neocloud GPU capacity, or higher-level data tooling

Source: Synergy Research Group, Q4 2025 cloud infrastructure market update (published Feb. 2026).

Six technologies are defining the cloud's current phase

The cloud is no longer just rented infrastructure; it is an operating model for software, data, and AI.

Containers + Kubernetes

The default control plane for portable distributed applications.

Serverless

Event-driven execution without provisioning servers explicitly.

Lakehouse / serverless analytics

Decoupled storage and compute unify data engineering, BI, and AI.

AI accelerators

GPUs, TPUs, and custom silicon now shape cloud cost/performance.

Hybrid + edge control

Policy and orchestration increasingly span data center, cloud, and edge.

Confidential computing

Sensitive workloads can protect data even while it is actively in use.

Today's competition is increasingly about who can offer the best software abstractions, data platforms, and AI infrastructure — not just the most virtual machines.

In practice, cloud is now ordinary infrastructure for extraordinary scale

Four examples show how cloud computing matters in media, finance, consumer analytics, and science.

Media
Netflix

Global streaming for 280M+ members in more than 190 countries on AWS infrastructure.

Finance
Capital One

Exited 8 data centers and improved disaster recovery and digital delivery speed.

Consumer data/ML
Spotify

Uses cloud data infrastructure to reduce operational burden and focus teams on insights and ML.

Science
Broad Institute

Cloud-based genomics workflows enabled roughly 4x faster genome analysis and global collaboration.

These are not edge cases. Cloud is now the default substrate for digital businesses, platform-native firms, and increasingly for science and public-interest research.

Examples adapted from AWS and Google Cloud customer case studies for Netflix, Capital One, Spotify, and Broad Institute.

PART III

Why Cloud Matters for Data Science

For data science, cloud computing changes not only scale but also collaboration, experimentation, and deployment.

The crucial change is that analysis, engineering, and production can now share the same platform.

For data science, the cloud is not just cheaper infrastructure

Its deeper value is that it compresses the distance between data, experimentation, collaboration, and production.

Scale

Object storage and distributed compute make it normal to work with data far beyond the limits of a single laptop or workstation.

Speed

Teams can provision notebooks, clusters, and GPUs on demand rather than waiting for local infrastructure.

Elastic compute + shared storage + managed platforms

Collaboration

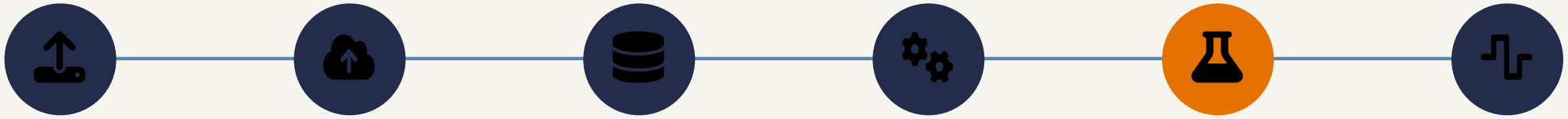
Shared datasets, pipelines, registries, and permissions make it easier to coordinate work across roles and teams.

Production

The same platform that supports exploration can also host training, deployment, monitoring, and governance.

A cloud-native data science workflow

The same cloud environment can support ingestion, storage, transformation, training, deployment, and monitoring.



Sources
APIs, logs, apps, sensors

Ingest
batch or streaming

Store
object storage / lakehouse

Transform
Spark, SQL, dbt, features

Train
experiments, notebooks, GPUs

Deploy + monitor
inference, drift, lineage

Why the cloud helps: storage and compute are separated, so each stage can scale independently; managed services also reduce the operational burden of moving work from prototype to production.

The cloud removes friction; it does not remove discipline

A mature data science organization treats governance, cost, reproducibility, and portability as design problems.

Cost sprawl

What strong teams do

Good teams use FinOps, quotas, storage lifecycle policies, and careful workload sizing rather than assuming elasticity is free.

Governance + privacy

What strong teams do

Access control, encryption, audit trails, and sometimes confidential computing are essential for sensitive data.

Reproducibility + drift

What strong teams do

Version data, code, features, environments, and monitor models after deployment rather than treating training as the finish line.

Lock-in + portability

What strong teams do

Use open formats, containers, and clear architecture boundaries when convenience creates strategic dependence.

The hardest cloud problems are often managerial and architectural rather than purely technical.

Three takeaways

1	**Historical**	Cloud computing is the mature form of an older utility-computing vision — now made practical by software-defined control, internet-scale networks, and hyperscale operations.
2	**Strategic**	The current cloud market is shaped not only by provider size, but by higher-level data platforms, AI infrastructure, and hybrid control planes.
3	**Data science**	For data science, cloud matters because it makes large-scale experimentation, collaboration, and production deployment practical on a shared platform.